

LESSON 7.3

CONSTRUCTING CIRCLE GRAPHS



LESSON INTRODUCTION

MA.7.DP.1.4 Use proportional reasoning to construct, display, and interpret data in circle graphs.

LEARNING TARGETS

- I can construct a circle graph given categorical data.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.9.DP.1.1 Given a set of data, select an appropriate method to represent the data depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

ESSENTIAL QUESTIONS

- How can you use given data to determine the angles and percentages the data represents in a circle graph?
- How do you construct a circle graph?

LESSON NARRATIVE

This lesson positions students to apply what they learned in the previous two lessons to construct circle graphs. Students determine what data can be displayed in a circle graph and calculate the percentages and angle measures of sectors needed to construct circle graphs.

COMPONENT SUMMARY

7.3.1 WARM-UP (~5 MIN.)

Determine whether a circle graph accurately represents data

7.3.3 COLLABORATION (15 MIN.)

Collaboratively calculate percentages and angles to construct circle graphs

7.3.5 WRAP-UP (~10 MIN.)

Demonstrate mastery of lesson learning goals

7.3.2 GUIDED INSTRUCTION (10 MIN.)

Determine what types of data can be displayed using a circle graph

7.3.4 YOUR TURN! (10 MIN.)

Practice constructing a circle graph from data

RESOURCES

GLOSSARY

Key Terms:

- Circle graph

Additional Terms:

- N/A

MATERIALS

- Protractors
- Straightedges

ADDITIONAL PREPARATION

- Consider having copies of large circle templates available for students who struggle with constructing circle graphs on the small circles in the student workbooks.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not realize they need to measure the next consecutive angle in the circle graph by measuring from the last ray of the previous angle.
- Students may misread the protractor scale by not consistently using either the top or bottom scale when measuring.

THINGS TO CONSIDER

- Students are asked to round percentages and angle measures to the nearest whole number. Discuss why this may result in the sum of the percentages not being 100% and the angles not having a sum of 360° . Consider if you would like students to round to a different place value in this lesson.
- Allowing students to use calculators during the lesson will help students focus on the discovery and process of finding central angles and percentages when constructing circle graphs.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

Consider using this routine with question 1 in 7.3.2 Guided Instruction. Ask students, “What do you notice?” Record all students’ responses. Avoid praising, restating, clarifying, or asking questions. Then ask students, “What are you wondering?” After recording responses, ask students, “Is there anything you need clarified?” If you or the students have questions about any items, ask the students who shared them to provide clarification.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.6.1 Reasonableness

Students have opportunities to construct and check the accuracy of circle graphs that represent real-world data throughout the lesson. Remind students to informally check their answers by asking themselves if they make sense and formally check their answers by determining if the sum of percentages equal 100% and sums of angle measures equal 360° .

DIFFERENTIATE INSTRUCTION

7.3.3 Collaboration provides an auditory entry point as students discuss how to construct a circle graph with their partners. The circle graphs provide visual entry points and using a protractor to construct the circle graphs provides a kinesthetic entry point.

Additional differentiation opportunities are denoted throughout the lesson guide.

7.3.1 WARM-UP: DATA DISPLAY

ENRICHMENT

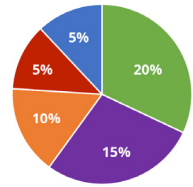
Consider asking students the following:

- What should the percentages in a circle graph add up to?
- What percentage of the circle does the green section appear to be?



7.3.1 Warm-Up: Data Display

Consider the circle graph.



1. What do you notice about the following graph?
The percentages do not add up to 100%.
2. Explain whether or not this is a valid display of data.
This is not a valid display of data because a whole circle is supposed to equal 100% and this graph on adds up to 55%.



7.3.2 Guided Instruction: Circle Graphs



A **circle graph**, also called a pie chart, is a visual display of categorical data. The whole set of data is represented by the circle and its interior. The categories are represented by fractional parts of the circle.

- The survey data in the table shows data collected in 2019 on the use of different technologies in the United States.

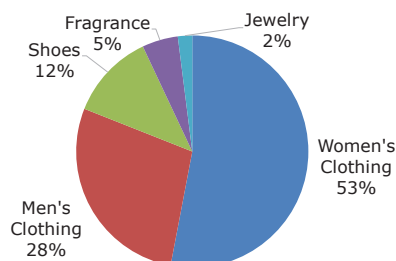
Technology	% of People Who Use It
Internet	86%
Cell Phone	96%
Tablet/Computer	88%

- How could someone using more than one of the technologies in the survey affect the percentages?
If someone uses more than one technology, the percentages will add up to more than 100%.
 - Discuss with your partner whether the data in the table can be displayed in a circle graph. Summarize your discussion.
The data from the table could not be displayed in a circle graph because the percentages do not add up to 100%. When the percentages are converted to fractions, their sum is greater than 1.
- A store collected data on the percentage of sales that each department generated for the store's total revenue over three months.

Department	% of Sales Generated
Women's Clothing	53%
Men's Clothing	28%
Shoes	12%
Fragrance	5%
Jewelry	2%

- With your partner, compare the department store data to the technologies data. Describe any similarities or differences you notice.
Answers will vary. The department store data is different from the technologies data because the percentages add up to 100%. The percentages from the technology data table are higher than the percentages in the department store data table.
 - Discuss with your partner why a circle graph is a good display for the department store data.
A circle graph is a good display for department store data because the percentages add up to 100%. Each department in a store can be represented by a sector of the circle graph, and the sector will show a good visual for the portion of sales generated by each department.
- The circle graph displays a store's sales by department over a three-month period.

Sales for October - December 2019



- Describe the information that was included for each sector.
Each sector shows what percentages of all the sales each department accounts for over the 3 month period.

7.3.2 GUIDED INSTRUCTION: CIRCLE GRAPHS

TEACHER MOVES

Notice and Wonder

Ask students, "What do you notice?" Record all students' responses. Avoid praising, restating, clarifying, or asking questions. Then ask students, "What are you wondering?" After recording responses, ask students, "Is there anything you need clarified?" If you or the students have questions about any items, ask the students who shared them to provide clarification.

QUESTIONING

As an extension, consider having students interview a manager at a local department store to gather their sales data. Then, have the students present their findings in tables and circle graphs.

7.3.2 GUIDED INSTRUCTION: CIRCLE GRAPHS (CONTINUED)

TEACHER MOVES

Students are not expected to write a response to question 3c, though they can if they choose to. Rather, use this discussion point as an opportunity to listen to students' thinking and address any misconceptions before they move on to question 3d.

- b. Which department had the largest percentage of sales?

Women's Clothing

- c. Discuss with your partner how to find the angle measure for each sector.

You can find the angle of measure for each sector by converting the percentage to a decimal and multiplying by 360° .

- d. Find the corresponding angle measure for each department. Round to the nearest whole number.

Department	Angle Measure ($^\circ$)
Women's Clothing	191 $^\circ$
Men's Clothing	101 $^\circ$
Shoes	43 $^\circ$
Fragrance	18 $^\circ$
Jewelry	7 $^\circ$

- e. What is the sum of all five calculated angle measures?

360 $^\circ$

- f. Use your protractor to compare your calculated values with the degrees of each sector in the actual circle graph. Then, discuss with your partner how they compare.

Some of the angles measure exactly the same as what I calculated, while others are slightly off due to rounding.



7.3.3 Collaboration: Constructing a Circle Graph

Complete the following with your partner.

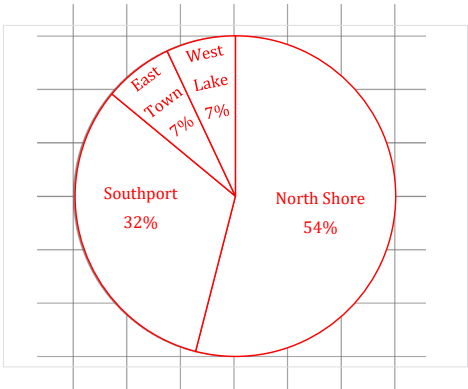
1. A travel agency sent out a survey to find out which local airports people preferred. The travel agency received 380 responses and compiled the data into the table shown.

Airport	Number of People Who Selected the Airport
Northshore	206
Southport	122
East Town	26
West Lane	26

- a. Use the data to complete the table. Round percentage to the nearest whole number and angle measure to the nearest tenth.

Airport	airport choice total responses	Percentage	Angle Measure (°)
Northshore	$\frac{206}{380}$	54%	194.4°
Southport	$\frac{122}{380}$	32%	115.2°
East Town	$\frac{26}{380}$	7%	25.2°
West Lane	$\frac{26}{380}$	7%	25.2°

- b. Use the data in the table and a protractor to construct a circle graph.



7.3.3 COLLABORATION: CONSTRUCTING A CIRCLE GRAPH

DIFFERENTIATE INSTRUCTION

Consider assigning one data set to each student partnership and then having pairs with different data sets share their work with each other. The second data set provides the percentages (less work). Have all students complete question 2b based on the graph that they constructed.

ENRICHMENT

For question 1a, consider asking students the following:

- When constructing your graph, which angle measure will you draw in first? Why?
- After drawing in your first angle, how will you continue?

7.3.3 COLLABORATION: CONSTRUCTING A CIRCLE GRAPH (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider providing students with a larger circle template for constructing their circle graph. Students may also need a straightedge to create the sectors.

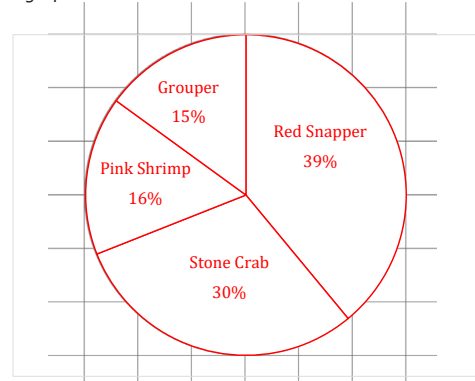
DIFFERENTIATE INSTRUCTION

Students may struggle writing a response to question 2b. Consider providing students with sentence stems for each step they took.

2. A restaurant is developing a new menu, so they decide to survey people's favorite local seafood. The results of the survey are shown in the table.

Favorite Seafood	Percentage Preferred
Grouper	15%
Red Snapper	39%
Stone Crab	30%
Pink Shrimp	16%

- a. Use the data and a protractor to construct a circle graph.



- b. Summarize the process you and your partner used to construct the circle graph.

To create the circle graph, I converted the percentages to decimals and multiplied each decimal by 360 to get the angle measure. Then I used my protractor to draw the largest angle first and then connected the smaller angles to until they formed a circle.

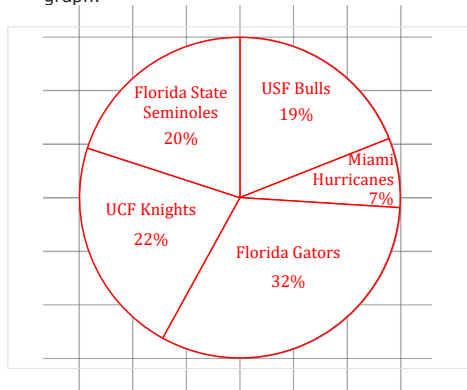


7.3.4 Your Turn!

- The Pasco Middle School Pirates are conducting a survey to determine their students' favorite Florida sports team name. The results of the survey are shown in the table.

Florida Team Name	Percentage Preferred
USF Bulls	19%
Miami Hurricanes	7%
Florida Gators	32%
UCF Knights	22%
Florida State Seminoles	20%

Use the data and a protractor to construct a circle graph.



7.3.4 YOUR TURN!

QUESTIONING

Consider having students conduct a similar survey in your school. Then, the students could represent the results in a table and in a circle graph.



7.3.5 Wrap-Up: Ticket Out the Door

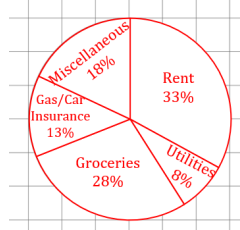
- The Rudolph Family's top five monthly expenses are as follows:

- Rent \$950.00
- Utilities \$236.00
- Groceries \$800.00
- Gas/Car Insurance \$356.50
- Miscellaneous \$500.00

- Complete the table. Round each percentage to the nearest whole number and angle measure to the nearest tenth.

Expense	expense category total expenses	Percentage	Angle Measure (°)
Rent	$\frac{950}{2842.50}$	33%	118.8°
Utilities	$\frac{236}{2842.50}$	8%	28.8°
Groceries	$\frac{800}{2842.50}$	28%	100.8°
Gas/Car Insurance	$\frac{356.50}{2842.50}$	13%	46.8°
Miscellaneous	$\frac{500}{2842.50}$	18%	64.8°

- Construct the corresponding circle graph.



7.3.5 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

For students who might struggle to complete this Wrap-Up in the time allotted, consider filling in some of the information in the table to reduce the calculations needed before constructing the circle graph.

The Wrap-Up could also be split, where the students complete the table in this lesson and construct the circle graph either for homework or in place of the Warm-Up in the next lesson.